(1) EM Spectrum

Ranges	γ	X	UV	vis	IR	micro	radio
Energy (E)	↑						↓
Frequency (ν)	↑						↓
Wavelength (λ)	↓						↑
Wavelength (λ)	<0.01	0.01–10	10–400	400–700	700–1e6	1e6–9	>1e9

- violet · indigo · blue · green · yellow · orange · red

(2) $c = \lambda\nu$ (Speed of light = wavelength × frequency)

c	ν (Hz)	kHz	MHz	GHz
λ (m)	2.9979e8 (m/s)	2.9979e5 (m·kHz)	2.9979e2 (m·MHz)	2.9979e-1 (m·GHz)
nm	2.9979e17 (nm·Hz)	2.9979e14 (nm·kHz)	2.9979e11 (nm·MHz)	2.9979e8 (nm·GHz)

(3) $E = h\nu = h(c/\lambda)$ (Energy = Planck × frequency)

hc, h	λ (m)	λ (nm)	ν (Hz)
E (J / photon)	1.9864e-25 (J·m)	1.9864e-16 (J·nm)	6.626e-34 (J·s)
J / mol	1.1962e-1 (J·m/mol)	1.1962e8 (J·nm/mol)	3.9902e-10 (J·s/mol)
kJ / mol	1.1962e-4 (kJ·m/mol)	1.1962e5 (kJ·nm/mol)	3.9902e-13 (kJ·s/mol)

- Grams → mol → ×6.022e23 for photon **count**

(4) Quantum Principles

Maxwell	Light is electromagnetic radiation
Planck	Energy emitted only in fixed amounts (quanta) (black-body radiation)
Einstein	Different light has different amounts of fixed energy (photons) (photoelectric effect)
Bohr	Electron quanta (H atom line spectra); Rydberg
De Broglie	Electrons are particle and wave (5)
Schrodinger	Wave equation (energy and spatial prob. of e-) (n, l, m_l, m_s)
Heisenberg	Position and velocity not both knowable

- $E = h\nu = h(c/\lambda) = R(1/n_f^2 - 1/n_i^2)$
- $n_i = \sqrt{1/\left(1/n_f^2 - hc/(R\lambda)\right)}$; $n_f = \sqrt{1/\left(1/n_i^2 + hc/(R\lambda)\right)}$

Variable	Value	Variable	Value
h	6.626e-34 J·s	h/R	3.0394e-16 s
c	2.9979e8 m/s	hc	1.9864e-25 J·m
R	2.18e-18 J	hc/R	9.1120e-8 m

(5) $\lambda = h/(m \cdot u)$ (De Broglie)

- 1 mm = 1e-3 m, 1 nm = 1e-9 m
- e- (9.11e-31 kg), p+ (1.6726e-27 kg), n (1.6749e−27 kg)

h	u (m/s)	km/s
m (kg)	6.626e-34 (J·s)	6.626e-37 (kJ·s)
g	6.626e-31 (mJ·s)	6.626e-34 (J·s)
amu	3.9892e-7 (J·s·amu/kg)	3.9892e-10 (kJ·s·amu/kg)

(6) Electron Configuration (Aufbau, Pauli, Hund)

- 1s 2s 2p 3s 3p 4s 3d 4p 5s 4d 5p 6s 4f 5d 6p 7s 5f 6d 7p
- Round-robin sublevel before double occupaton (Hund)
- Two e- per orbital (Pauli's exclusion)
- n^2 orbitals, $2n^2$ e- per shell

Symbol	Name	Value
n	**Principal energy level** (shell)	(1, 2, 3, ...)
l	**Sublevel** = subshell (shape)	(0, ..., n - 1)
m_l	**Orbital** (orientations)	(-l, ..., l)
m_s	**Spin** (2 per orbital)	$\pm 1/2$

Sublevel	l	m_l	# orbitals	# e-
s	0	0	1	2
p	1	±1	3	6
d	2	±2	5	10
f	3	±3	7	14

Element	EC	Element	EC
Cr	$[Ar]4s^1 3d^5$	Ag	$[Kr]5s^1 4d^{10}$
Mo	$[Kr]5s^1 4d^5$	Au	$[Xe]6s^1 4f^{14} 5d^{10}$
Cu	$[Ar]4s^1 3d^{10}$	Pd	$[Kr]4d^{10}$ (no 5s)

(7-8) Reading EC, Anion, Cation

EC → element	add every superscript to get Z
Paramagnetic	at least 1 unpaired val. e- (else dia.)
Ground state	follows Aufbau (else excited)
Valence electrons	highest / outer shell (else core)
Anion	count val. e- needed until noble gas
Cation	remove val. e- until noble gas
Tx metal	remove s before d
Isoelectric	same EC (e.g. Na+, Al3+, F-, N3-, Ne)

(9-10) Penetration, Shielding, Z_{eff}, Size

$$Z_{eff} \approx Z - (\text{core } e^-)$$

- ↑ penetration (core electrons) = ↓ shielding = ↓ energy
- **Energy** s < p < d < f (same shell)
- **Core** e- shield valence e-
- **Atomic size** (cation < neutral < anion)

(11-13) Trends

	Trend	Top	Bot	Left	Right
(11) Atomic size (radius)	↙ (Cs) (f > p)	↓	↑	↑	↓
(12) Ionization energy (IE)	↗ (He)	↑	↓	↓	↑
(13) Electron affinity (EA)	↗ (Cl)	↑	↓	↓	↑
Z_{eff} (+ ch. toward nucl.)	↘	↓	↑	↓	↑
Metallic	↙ (Fr)	↓	↑	↑	↓
Nonmetallic	↗ (F)	↑	↓	↓	↑
Electronegativity (EN)	↗ (F)	↑	↓	↓	↑

- EA equation: X + e- = X-

(14) Families & Diagonal Rule

Family	Name	Reactivity	e-	Charge
1A	alkali metals	(downward)	lose 1	1+
2A	alkaline earth	less than 1A	lose 2	2+
7A	halogens	(upward)	gain 1	1−
8A	noble gases	**inert**	—	0

- **Diagonal rule** (Li~Mg, Be~Al, B~Si)
- **Lanthanide** (58–71), **Actinide** (90–103)
- **Diatomic** (H_2, N_2, O_2, F_2, Cl_2, Br_2 (l), I_2 (s))
- **Oxides** (basic = metal, acidic = nonmetal)
- **Family reactions**
  - $2K(s) + 2H_2O(l) \rightarrow 2KOH(aq) + H_2(g)$
  - $H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$

(15) Lattice Energy

$$U \propto \frac{q_1 q_2}{r_1 + r_2}$$

- Energy **released** (exo.) forming ionic crystal (from gas)
  - (Lattice *dissociation* energy = endothermic)
- max U: 1. ↑ Q+/Q- (ionic charge), 2. ↓ AS (rad.)

Trend	Bond	Enthalpy	Exception
EA	Form	Exothermic	Endo: Be, Mg, noble, EA2 (usually)
IE	Break	Endothermic	--

(16-17) Electronegativity, Bond Polarity

$$\Delta EN = |EN_A - EN_B|$$

- EN = e- attraction in cov. bond.
- ↑ EN = ↑ bond = ↑ polar = shorter (not always)
- triple > double > single (ranking)

ΔEN	Label
0	pure non cov.
0.1 – 0.5	slightly pol. cov.
0.5 – 1.6	pol. cov.
> 1.6	ionic

(18) Formal Charge & Resonance

$$FC = (\text{valence } e^-) - [\text{nonbonding } e^- + \#\,\text{bonds}]$$

1. Overall 0 charge is better
2. Else −1 on the **more electronegative** atom

- Equivalent resonance → real bonds are **average** (e.g. 1½)

(19) Lewis Structures (Kelter)

1. Sum valence e- (including +/- for ions)
2. Sum "wants" (octet)
   - H (2), He (2), - Be (4), Al (6), B (6)
   - 3p or below (>8)
3. Sum bonds = (2. - 1.) / 2 (round down)
4. Central atom = min EN (never H)
5. Remaining e- = 1. outside, central,
   - Consider double/trip. bond for central octet / nonzero FC
   - Ions go in brackets with the charge outside
6. FC: overall 0, or 1- on max EN element

(20) Photoelectric Effect ($h\nu$ - binding / work)

$$KE = h\nu - \Phi = h(c/\lambda) - \Phi = \frac{1}{2}m_e u^2$$

Variable	Value
h	6.626e-34 J·s
m_e	9.11e-31 kg
hc	1.9864e-25 J·m
$2hc/m_e$	4.3609e5 m^3/s^2

Have	Want	Use
λ, λ_{thr}	u	$u = \sqrt{(2hc/m_e)(1/\lambda - 1/\lambda_{thr})}$
Φ	λ_{thr} (max ej. λ)	$\lambda_{thr} = hc/\Phi$
ν_{thr} or λ_{thr}	Φ, Φ / mol	$\Phi = h\nu_{thr} = h(c/\lambda_{thr})$, n.b. Φ × N_A (/1e3 for kJ/mol)
Φ, λ	u	1. $KE = h(c/\lambda) - \Phi$ 2. $u = \sqrt{2\,KE/m_e}$

(21) Heisenberg Uncertainty

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi} = 5.2728\text{e}{-35}\ \text{J·s}, \quad \Delta p = m\Delta u$$

1. $\Delta p = m \cdot \Delta u$ (kg, m/s)
2. $\Delta x \geq h/(4\pi \cdot \Delta p)$ (**minimum** uncertainty)

(22) Born–Haber Cycle

$$\Delta H^{\circ}{}_f = \Delta H_{sub} + \Sigma IE + \frac{1}{2}BE + EA + U$$

1. List every step with its sign: sublimation (+), IE (+), bond dissociation (+), EA (usually −), U (−)
2. **Halve** the bond dissociation if you need only one atom
3. Chain: elements → gaseous atoms → gaseous ions → solid
4. Set the chain equal to $\Delta H^{\circ}{}_f$
5. Solve for the single unknown

- Use IE_2 as well when the cation is 2+
- U runs ions(g) → solid, so it comes out **negative**; report |U| if asked for a positive lattice energy

(23) Bond Enthalpy

$$\Delta H_{rxn} = \Sigma\Delta H(\text{bonds broken}) - \Sigma\Delta H(\text{bonds formed})$$

1. List bonds broken (reactants) and formed (products), using the balanced coefficients
2. Σ BE(broken), always positive
3. Σ BE(formed), subtracted
4. ΔH = Σ broken − Σ formed

- Bonds unchanged on both sides cancel — skip them
- Negative ΔH = exothermic
- For "heat released by g of X," convert g → mol and scale

Average Bond Energies (kJ/mol)

Bond	kJ/mol	Bond	kJ/mol	Bond	kJ/mol
H-H	436	N-N	163	Br-F	237
H-C	414	N=N	418	Br-Cl	218
H-N	389	N≡N	946	Br-Br	193
H-O	464	N-O	222	I-Cl	208
H-S	368	N=O	590	I-Br	175
H-F	565	N-F	272	I-I	151
H-Cl	431	N-Cl	200	Si-H	323
H-Br	364	N-Br	243	Si-Si	226
H-I	297	N-I	159	Si-C	301
C-C	347	O-O	142	S-O	265
C=C	611	O=O	498	Si=O	368
C≡C	837	O-F	190	S=O	523
C-N	305	O-Cl	203	Si-Cl	464
C=N	615	O-I	234	S=S	418
C≡N	891	F-F	159	S-F	327
C-O	360	Cl-F	253	S-Cl	253
C=O	736 (799 in CO_2)	Cl-Cl	243	S-Br	218
C≡O	1072			S-S	266
C-Cl	339				